

Grade 10 lessons

Geometry: angles, similarity, trig and circles

IN THIS BOOK

1. Angles
2. Similar triangles
3. Right-triangle trig
4. Circles and solids
5. Coordinate geometry

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 10 Kahoot and Blooket sets, and one stop on the Grade 10 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Angles

THE BIG IDEA

Angles follow a few fixed rules. Knowing them lets you find a missing angle without measuring.

HOW TO DO IT

1. The three angles in a triangle add to 180° .
2. Angles on a straight line add to 180° . They are called supplementary.
3. Angles in a right-angle corner add to 90° . They are called complementary.
4. When two lines cross, the angles opposite each other are equal.
5. The angles inside a polygon with n sides add to $(n - 2) \times 180^\circ$.

WORKED EXAMPLES

A triangle has angles of 50° and 60° . Find the third.

$$180 - 50 - 60 = 70^\circ.$$

What is each angle of a regular hexagon?

$$\text{Sum: } (6 - 2) \times 180 = 720^\circ. \text{ Six equal angles: } 720 \div 6 = 120^\circ.$$

WATCH OUT

Supplementary adds to 180 (a straight line). Complementary adds to 90 (a corner). Do not mix them up.

WORDS TO KNOW

regular polygon

a shape with all sides and all angles equal

vertical angles

the opposite angles where two lines cross

Similar triangles

THE BIG IDEA

Similar shapes have the same angles and the same shape but a different size. Every length is multiplied by the same number, called the scale factor.

HOW TO DO IT

1. Match up the sides that correspond.
2. Divide a big side by its matching small side to find the scale factor.
3. Multiply any other small side by the scale factor to get its matching big side.
4. Areas are multiplied by the scale factor squared.

WORKED EXAMPLES

A triangle has sides 3, 4, 5. A similar one has 9 matching the 3. Find its other sides.

Scale factor: $9 \div 3 = 3$. The other sides are $4 \times 3 = 12$ and $5 \times 3 = 15$.

A 6 ft pole casts a 4 ft shadow. A tree casts a 20 ft shadow. How tall is the tree?

The tree shadow is 5 times as long.
So the tree is $6 \times 5 = 30$ ft.

WATCH OUT

Match the right sides before you divide. Lengths use the scale factor. Areas use the scale factor squared.

WORDS TO KNOW

similar

same shape, different size

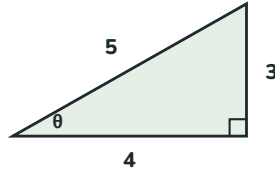
scale factor

the number every length is multiplied by

Right-triangle trig

THE BIG IDEA

In a right triangle, the ratios between the sides depend only on the angle. Those ratios are called sine, cosine and tangent.



From angle θ : opposite is 3, adjacent is 4, hypotenuse is 5.

HOW TO DO IT

1. Find the hypotenuse: it is across from the right angle.
2. From your angle θ , the opposite side is across from it and the adjacent side is next to it.
3. Use SOH-CAH-TOA: $\sin \theta = \text{opposite} \div \text{hypotenuse}$, $\cos \theta = \text{adjacent} \div \text{hypotenuse}$, $\tan \theta = \text{opposite} \div \text{adjacent}$.
4. Special values to know: $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\tan 45^\circ = 1$.

WORKED EXAMPLES

In the triangle shown, find $\sin \theta$, $\cos \theta$ and $\tan \theta$.

$$\sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}.$$

$\sin \theta = \frac{3}{5}$ and the hypotenuse is

20. Find the opposite side.

$$\text{Opposite} = \text{hypotenuse} \times \sin \theta = 20 \times \frac{3}{5} = 12.$$

WATCH OUT

Opposite and adjacent depend on which angle you are looking from. The hypotenuse never changes.

WORDS TO KNOW

hypotenuse

the longest side, across from the right angle

adjacent

the side next to the angle

Circles and solids

THE BIG IDEA

Circles and round solids all use π and the radius. A part of a circle takes the same fraction of the area as its angle takes of 360° .

HOW TO DO IT

1. Circle: area = πr^2 , circumference = $2\pi r$.
2. Sector (a slice): area = $(\text{angle} \div 360) \times \pi r^2$.
3. Arc (part of the edge): length = $(\text{angle} \div 360) \times 2\pi r$.
4. Cylinder: volume = $\pi r^2 h$. A cone with the same base and height is one third of that.
5. Sphere: volume = $\frac{4}{3}\pi r^3$.

WORKED EXAMPLES

A sector has a 90° angle and radius 4. Find its area.

$90 \div 360 = \frac{1}{4}$. The whole circle is 16π . A quarter of that is 4π .

A cylinder has radius 3 and height 5. Find its volume.

$\pi \times 3^2 \times 5 = \pi \times 9 \times 5 = 45\pi$.

WATCH OUT

Area uses r squared. Circumference uses $2r$. Leave π in the answer unless you are told to round.

WORDS TO KNOW

sector

a slice of a circle

arc

part of the edge of a circle

Coordinate geometry

THE BIG IDEA

On a coordinate grid you can find distances, midpoints and slopes using only the coordinates of the points.

HOW TO DO IT

1. Distance: find the run and the rise between the points, then use $\sqrt{(\text{run}^2 + \text{rise}^2)}$.
2. Midpoint: average the x values and average the y values.
3. Parallel lines have the same slope.
4. Perpendicular lines have slopes that are negative reciprocals: flip the fraction and change the sign.
5. A circle $(x - h)^2 + (y - k)^2 = r^2$ has center (h, k) and radius r .

WORKED EXAMPLES

Find the distance from (1, 2) to (4, 6).

Run: 3. Rise: 4. $\sqrt{9 + 16} = \sqrt{25} = 5$.

Find the midpoint of (2, 4) and (6, 10).

x: $(2 + 6) \div 2 = 4$. y: $(4 + 10) \div 2 = 7$.
The midpoint is (4, 7).

WATCH OUT

In the circle equation the signs are flipped. $(x - 3)^2 + (y + 1)^2 = 25$ has center (3, -1), and the radius is 5, not 25.

WORDS TO KNOW

midpoint

the point halfway between two points

reciprocal

a fraction turned upside down